

Sammer Hussain

Junior AI Software Engineer

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Summary

Junior AI Software Engineer with hands-on experience building LLM-powered chatbots, RAG pipelines, and full-stack AI applications, currently at Decotechs. Comfortable working across the full stack, from embeddings and vector databases to APIs and user-facing interfaces, with a focus on turning AI prototypes into systems people can actually use.

Experience

Junior AI Software Engineer — Decotechs, Karachi (Onsite) Feb 2026 – Present

- Built and deployed LLM chatbots and RAG pipelines using LangChain, LangGraph, OpenAI APIs, and Ollama for context-aware knowledge retrieval.
- Built RESTful APIs with FastAPI, Node.js, and Express; built responsive frontends with React.js and Streamlit.
- Automated internal workflows using n8n and Playwright for browser-based process automation.
- Built the Medication Administration Record (MAR) module for a MERN-stack medical data entry system, displaying medicines on daily, weekly, and monthly schedules and logging status (taken, refused, unable to take).
- Leading development of an LLM-powered text-to-CAD 3D model generation feature, currently moving toward production deployment.

AI Engineer Intern — CoreTech Innovations, Hyderabad (Remote) Aug – Sep 2025

- Built and evaluated classification models (Logistic Regression, Random Forest with GridSearchCV tuning) on a benchmark survival-prediction dataset, improving accuracy from 80% to 82%.
- Built an SMS spam-detection system comparing 10+ ML algorithms; a voting-classifier ensemble reached 98.2% accuracy and 99.2% precision.
- Built a multi-class service-inquiry classifier (5 categories, TF-IDF + Logistic Regression/Random Forest) for a smart service platform, achieving near-perfect accuracy on the evaluation set.
- Built and evaluated sentiment-analysis models using Python, Scikit-learn, and NLTK.

Education

B.S. Information Technology

Quaid-e-Awam University of Engineering, Sciences & Technology
Nawabshah, Pakistan
Dec 2021 – Dec 2025 CGPA: 3.57/4.00

Skills

AI / Agentic

[LangChain](#) [LangGraph](#) [AI Agents](#) [Multi-Agent System](#) [MCP](#) [RAG](#)
[Prompt Eng.](#) [HF Transformers](#) [Fine-Tuning](#) [Scikit-learn](#) [TensorFlow](#) [NLTK](#)

LLMs & Vector DBs

[GPT-4o](#) [Gemini](#) [Claude](#) [Ollama](#) [Pinecone](#) [ChromaDB](#)

Backend & Frontend

[Python](#) [FastAPI](#) [Flask](#) [Node.js](#) [Express.js](#) [REST APIs](#) [React.js](#) [Streamlit](#)
[JavaScript](#) [HTML/CSS](#)

Databases & DevOps

[MongoDB](#) [PostgreSQL](#) [MySQL](#) [SQLite](#) [n8n](#) [Playwright](#) [Docker](#) [AWS](#) [Git](#)

Certifications

- Google AI Essentials
- Introduction to Generative AI
- IBM Machine Learning
- Full Stack MEAN Developer

Projects

AI Resume-Based Job Matching Portal [Link](#) • Feb 2026

[React](#), [FastAPI](#), [Node.js/Express](#), [MongoDB](#), [Pinecone](#)

- Built a three-service platform (React frontend, Node/Express API, FastAPI AI microservice) that parses uploaded resumes (PDF/DOC/DOCX) and matches them to jobs.
- Designed a weighted relevance-scoring model combining experience match (50%), skill overlap (30%), and semantic similarity from sentence embeddings (20%), normalized to a 0–100 score.
- Used Sentence Transformers (all-MiniLM-L6-v2) with Pinecone for vector storage/search, and JWT authentication across user and admin roles.

FredCoach: AI Coaching Platform [Link](#) • Oct – Nov 2025

[React](#), [Node.js](#), [Express](#), [MongoDB](#)

- Built a subscription-based AI coaching platform offering 24/7 chatbot coaching, powered by GPT-4o with RAG (ChromaDB).
- Implemented Stripe subscription billing, JWT authentication, and an admin dashboard for customizing branding, prompts, and assistant behavior.
- Added multi-chat support so users can manage multiple ongoing conversations.

Content Examiner: AI Plagiarism Detection (FYP) [Link](#) • Feb – Jun 2025

[Flask](#), [React](#), [RoBERTa](#), [TF-IDF](#)

- Built a plagiarism-detection web app combining RoBERTa and TF-IDF, supporting PDF/DOCX/TXT uploads and direct document/text comparison.
- Added web-search integration to automatically surface related online sources, with reports exported to PDF and plagiarism visualized via pie charts.

Voice AI Agent: Patient Registration [Link](#) • Sep 2026

[LangGraph](#), [FastAPI](#), [Twilio](#), [React/shadcn](#)

- Built an end-to-end voice agent connecting a real phone number via Twilio ConversationRelay (Deepgram STT, ElevenLabs TTS) to a LangGraph conversational agent for patient registration, plus a REST API and dashboard.
- Implemented guard-rails in code: mandatory read-back confirmation before saving, duplicate-caller detection with DOB verification, and shared field validation between the voice agent and the API.
- Covered with 24 automated tests (validation rules, REST endpoints, the LangGraph agent, and an end-to-end WebSocket call flow).

Live Object Detection with YOLOv8 [Link](#) • Aug – Sep 2024

[Flask](#), [React](#), [OpenCV](#), [YOLOv8](#)

- Built a real-time object detection app on YOLOv8x, supporting both live webcam (MJPEG stream) and image-upload detection across 80+ COCO classes.
- Added monocular distance estimation, adjustable confidence thresholds, class filtering, and live FPS/processing-time metrics.